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Java Tools for the Palm OS®

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Agenda

- Why Java on Palm™ handhelds?
- Background on Java 2 Micro Edition (J2ME)
- PDA Profile
- Palm Java VM demos—Sun, IBM/OTI, esmertec, Emwerks

Why Java on Palm™ Handhelds?

- Enterprise and wireless developers already using Java on the server, on the desktop, and in conduits
- Desire for a high-level, object-oriented toolkit on the Palm OS®
- Security
- Processor independence
- Platform independence

Java 2 Micro Edition (J2ME)

- Version of the Java 2 platform targeted at consumer electronics and embedded devices
- Portability is achieved through configurations and profiles
- Configurations and profiles are developed through the Java Community Process (JCP)

J2ME Configurations

- Consists of language features, runtime, and VM specifications for a broad category of devices
- UI not defined in configurations
- Connected Limited Device Configuration (CLDC)
- Connected Device Configuration (CDC)

J2ME Profiles

- Specifies a set of APIs suitable to a specific type of device, e.g. mobile phone, PDA, web pads
- Built on top of a configuration
- Mobile Information Device Profile (MIDP)
- Personal Digital Assistant Profile (PDAP)
- Personal Profile (successor to Personal Java)

PDA Profile

- JSR 75 approved August 8, 2000
- Dan Podwall, Palm, Inc. is specification lead
- Expert group consists of: Esmertec, EyeShake, Stefan Haustein (kAWT), IBM/OTI, Michael Kroll (kAWT), Motorola, Nokia, RIM, Samsung, Sharp, Siemens, Sony, Sun, Symbian, Wind River, Zucotto Systems

Goal of the PDA Profile

- MIDP has some good APIs, but the UI is too limited for Palm™ handhelds and other PDA-like devices
- Need access to standard PDA features, such as address book and calendar information
- Use of an AWT subset provides good transfer of knowledge for J2SE developers
- Non-UI code can be portable between PDAP and MIDP
- Implementations of PDAP for Palm OS can use the native Palm UI

Scope of the PDA Profile

- User interface
- Data storage
- Networking
- Timers
- Calendar
- Address book
- Application packaging and lifecycle
- other TBD



Demo—Metrowerks CodeWarrior Palm Java RAD

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Developing Java Apps on Palm OS® Using IBM VisualAge Micro Edition

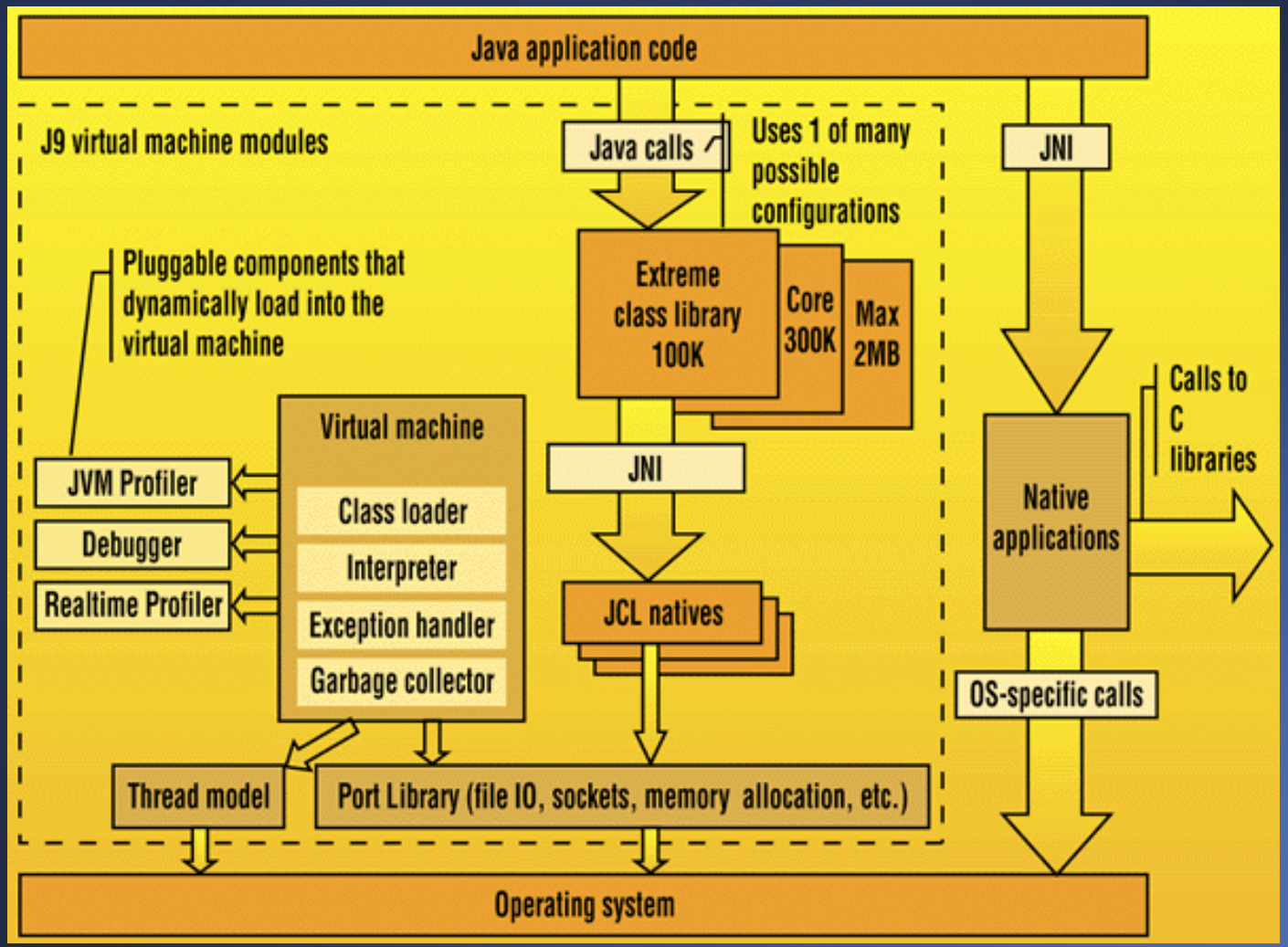
Chris Laffra
Object Technology International

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IBM Java Solution for the Palm OS®

- Free VisualAge Micro Edition Personal Edition IDE
  - See CD distributed at conference), or
  - Free download, no runtime fees (http://www.ibm.com/embedded)
- Distributed development team:
  - Everywhere: common IDE
  - Ottawa: VM + base class libraries
  - Amsterdam: Palm libs + samples + release build
  - Phoenix: Release building

J9, Common Code Base Strategy



Very Short Demo

- Show increasing level of abstraction:
  - Resources written with pilrc loaded in Java
  - Dynamic Palm OS forms
  - AWT (full Java, portable)

More Information?

- Visit us at the IBM booth
- Come to talk 605, Thursday, 1:45–2:45
- Come to the IBM lab, Thursday afternoon
- Ask someone to beam you the IBM Conference Scheduler

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esmertec, inc.

Luke Teyssier
Regional Director of
Technology North America

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esmertec Background

- Incorporated April 1999, Zurich, Switzerland
- Core Technology based on work starting in 1990
- US Office Open in April 2000
- Venture Capital Financed
- Rapidly Growing market presence

Jbed on Palm OS®

- Drop-in replacement for Sun KVM
- All code compiled before running
- Compiled Way Ahead of Time to native:
  - Pre-compiled fully linked native code on the host, or
  - Dynamic compile to native on the target, or
  - A combination (using classForName)
- Much faster than interpreted solutions, such as Sun KVM

Fully Pre-Compiled Java

- Java Source code compiled with Sun JDK 1.1.8 or compatible compiler
- Static Link:
  - Jbed brings together all bytecode files need for application, and links into a single .prc file
  - Download and run the .prc file just as if it were written in “C” (a standard palm application)

Dynamically Loaded Classes

- Dynamic Loading of Classes
  - bytecode stored ahead of time on handheld with bytecode to .prc tool
  - Can dynamically add classes with Class.forName
  - Database loader looks up from Palm OS files at runtime and loads, then compiles to native

Speed Comparison

- Jbed pre-compiled applications vs. Sun KVM apps
- Jbed CLDC approaches speed of C++
- Demonstration of pre-compiled vs. Sun KVM interpreted

Jbed and Graphics on Palm OS®

- Native Palm OS calls
- Spotlet class (except for beaming)
- KAWT on top of Spotlet
- MIDP—Mobile Information Device Profile
  - Shipping Beta by Jan 2001
- esmertec is on the PDA profile committee

Communication Protocols

- Support Socket communication through TCP
- Datagram Sockets (UDP)
- Serial Communication
- Server Sockets
- http://

Conclusion

- Jbed is a small, fast, and versatile solution for Java on Palm OS platforms
- Compatible with standards: CLDC, MIDP, etc.
- Allows calling of Palm OS functions
- Customer input drives our development
- New features are being added constantly


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Kada™ Mobile Application Platform

Shekar Mantha,
Founder and CTO

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Full-Function Java Support

- Kada is the first and only full-function implementation of Personal Java for Palm OS®.



History

- 2/00 Founded
- 9/00 Series A financing—JK&B Capital
- 6/97 Initiated development of Kada

Mobile Devices

- The mobile device market demands:
  - More sophisticated applications
  - Reliable connected and disconnected operation
  - Portability across operating platforms
- But you are probably frustrated by:
  - Unreliable, slow or non-existent connection services
  - Multiple operating environments
  - Portability problems caused by device constraints

Solution

- The Emwerks KADA Mobile Application Platform (MAP)
  - 100% compliant implementation of JVM and Java APIs
  - Uniform implementation for Palm OS, WinCE, EPOC and RIM platforms
  - Complete environment to run in connected, unconnected or intermittently connected modes
  - Seamless portability across PDAs, cellphones and wireless devices
  - FAST portability—new platform in eight weeks

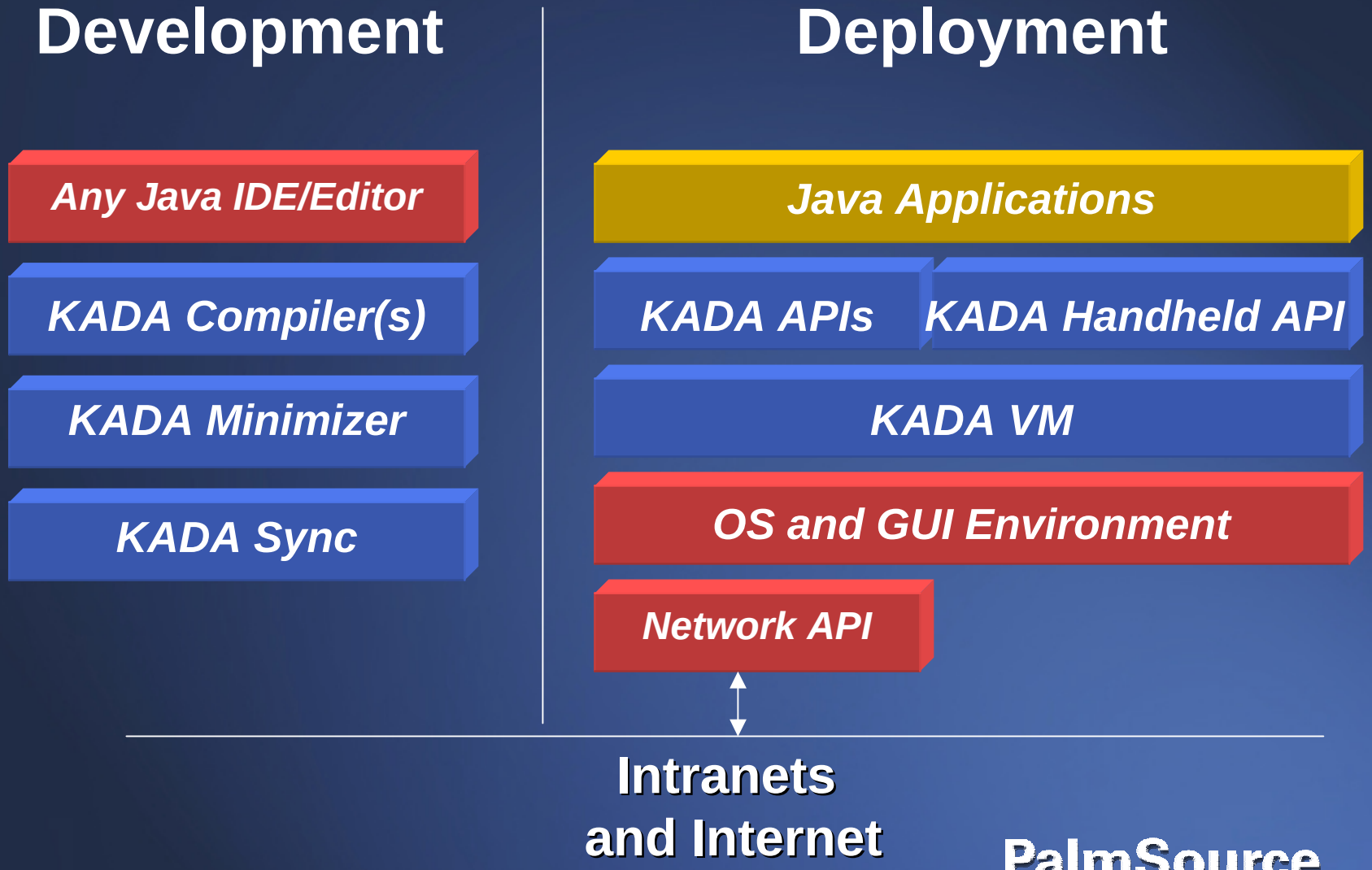
Kada Mobile Application Platform

Personal Java Applications



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Platform Architecture



Development Tools

- Kada works with any Java IDE
  - CodeWarrior, Kawa, Forte, VisualCafe, etc.
  - Remote (host-target) source level debugging (JPDA)
- Kada AOT Compiler
  - Tunes applications for speed by pre-compiling frequently executed methods
- Kada Minimizer
  - Reduces application size by eliminating unused classes, fields and methods

Development Tools

- Kada Sync
  - Enables easy application deployment onto handhelds

Deployment: KadaVM

- Footprint—160 KB to 384 KB on Palm OS
- Smart Class Loader
  - Static and dynamic class loading
  - Network, memory and file-system based loading
  - Pre-resolved class execution
- Adaptive JIT Compiler
  - 300% to 1400% speed improvement
  - Configurable to enable applications to work under memory constraints
- Configurable ROM-based execution

Deployment: Kada APIs

- API Packages460KB Footprint
  - lang, reflect, io, util, zip, net, text
  - awt, applet
  - sql, math
  - Coming soon: security, rmi and beans

- Handheld API Packages
  - Infrared, serial, alarms and database

JDK 1.1.8 Now – Java 2 in Q1 2001

Platform Support

- Palm OS 3.5
  - JDBC database support
  - 6 MB of Java heap and stack space
  - Multi-threading, AWT, Object Serialization, JNI, Floating Point and JIT Compiler
- Additional Platforms Supported
  - WinCE
  - EPOC 6.0
  - RIM

Contact Information

- Booth #227E
- www.emwerks.com
- 1-877-NOW-JAVA
- Early Access 1.0 Software January 2, 2000
- Full release February 11, 2000


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